



School segregation in the presence of student sorting and cream-skimming: Evidence from a school voucher reform

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ABSTRACT

This paper uses a reform to Chile's school choice system to study student socioeconomic segregation with a focus on student demand and school selectivity. The reform increases the subsidies that schools receive for low socioeconomic status students. I exploit this shock to schools' incentives to test for selection at admission based on students' socioeconomic characteristics. Schools respond to the new voucher by decreasing the level of cream-skimming. I incorporate these admission restrictions in a demand model to estimate parents' preferences for school and peer characteristics. I show that ignoring admission restrictions leads to underestimating poor parents' preferences for school quality. Counterfactual simulations show that preferences of high-SES parents for high-SES peers are one of the main drivers behind segregation as opposed to schools' selective behavior. This likely explains the unexpected increase in enrollment for schools that opted out of the reform and the ineffectiveness of the reform in reducing socioeconomic segregation across schools.

1. Introduction

Educational reforms in numerous countries introduce competition between schools by increasing parental choice via school vouchers.¹ In theory, increased competition between educational institutions should result in the provision of better school quality to attract students.² However, there is concern that school choice programs may increase social stratification in education systems and weaken public schools if higher-income students migrate to private voucher schools (Manski, 1992; Epple and Romano, 1998; Nechyba, 1999). Indeed, entry of private schools has been associated with stratification, consistent with private schools cream-skimming high-income students from the public sector (Hsieh and Urquiola, 2006; McEwan et al., 2008). Such increased segregation may be an important contributor to long-run inequality.

Studies on school desegregation plans in the late 1960s and 1970s have linked increased school segregation with increased criminal activity, lower educational attainment for minorities, and lower graduation rates (Guryan, 2004; Weiner et al., 2009; Billings et al., 2014).

This paper studies demand- and supply-side channels that contribute to socioeconomic segregation in school choice programs. There are several factors that affect school segregation: schools' selection processes, parents' choices, potential schooling fees, and residential segregation. Disentangling these mechanisms is challenging due to simultaneity. On the supply side, even in the absence of tuition, private schools may have incentives to select higher-income students to improve overall test results.³ On the demand side, the potential effects of school choice programs on segregation depend on parents' preferences for different

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¹ Chile, The Netherlands, Sweden, Denmark, New Zealand, India, Pakistan, Colombia, and the U.S have all implemented school choice programs of different scales.

² The case for educational vouchers and increased educational choice was initially made by Friedman (1962). Yet, empirical evidence does not show systematic effects in achievement or efficiency in either direction. Hoxby and Murarka (2009) study charter schools in New York City, and Hoxby and Rockoff (2004) study charter schools in Chicago finding modest gains. Rouse (1998) studies a voucher program in Milwaukee and finds no effects on reading, but significant effects on math. Lavy (2010) finds positive effects of a school choice program in Tel Aviv, both in terms of academic achievement and behavioral outcomes. Angrist et al. (2002) examine vouchers in Colombia, finding large positive effects. Muralidharan and Sundararaman (2015) analyze voucher experiments in India and find no effect on test scores, except for Hindi, but they also show that private schools spend much less than public schools. Generally, results depend on the context and design of the choice program and are heterogeneous across different groups. See recent review by Epple et al. (2017). Several papers have studied the Chilean voucher program implemented in 1981, finding no evidence that choice improved average educational outcomes (Hsieh and Urquiola, 2006; McEwan et al., 2008).

³ Numerous studies show that family income and parental education are the main factors explaining student achievement and standardized test results. Thus private schools may attract parents and students on the basis of superior average levels of test scores, but higher average test scores may be explained by sorting of self-selected high achievers, so schools may not be actually adding much value (Abdulkadiroglu et al., 2014; Cullen et al., 2006).

school characteristics and peer composition. Heterogeneous preferences across different socioeconomic groups may explain how parents sort across different schools. For instance, high-income parents may focus more on school quality, while low-income parents may focus more on convenience factors, such as distance.⁴ Furthermore, high correlation between socioeconomic status and test scores, make it difficult to disentangle whether parents care more about test scores or peer quality.⁵

I exploit a reform to the Chilean voucher system implemented in 2008 to study the relative importance of some of these mechanisms with a focus on school selectivity and demand heterogeneity. Chile is one of the few countries that has a nation-wide voucher program which makes it particularly suitable to study student sorting and segregation in educational markets. This reform changed the previous flat voucher (same per-student amount across schools) to a two-tier voucher based on students' socioeconomic status (SES), with participating schools receiving a larger voucher for low-SES students, with the requirement of eliminating any additional charges to eligible students. This change constitutes an exogenous shock to schools' incentives to select more low-SES students, uncorrelated with parents' preferences in the short term.

The empirical analysis to study student sorting is divided into three sections. First, a descriptive analysis of changes in enrollment, tuition, and segregation using school and year fixed effects. Second, an analysis of school admissions to examine changes in school selection following the reform. Third, a demand estimation to understand the drivers behind student sorting, considering the restrictions that some students face in light of the private school selectivity. I exploit the reform in two ways: first, to test for cream-skimming behavior in the group of participating schools in the analysis of school admissions. To this purpose, I look at changes within school in the lowest type of students admitted before and after the implementation of the program. Second, in the demand estimation, to identify preference parameters regarding peer characteristics through the induced variation in peer composition within schools.

The descriptive analysis provides evidence of two effects of the reform related to student sorting. First, I find a 10 percentage point increase in the probability of low-SES students enrolling in participating private subsidized schools (SEP schools). This can be partly explained by the elimination of fees for low-SES students in SEP schools, although more than half of these schools had no tuition before the reform. Second, my findings show high-SES student enrollment gradually increasing in private subsidized schools that opted out of the reform (non-SEP schools). This shift is less straightforward, as the new policy does not directly affect the subsidies received by non-participating schools. Nevertheless, competitive forces affect all schools and several school characteristics respond endogenously, in particular prices and peers. If participating schools increase their prices for high-SES students, this could explain the migration towards non-participating schools. In fact, evidence shows that prices are moving in the opposite direction, increasing in non-participating schools and decreasing in participating schools. Contrary to the goals behind the reform, socioeconomic segregation does not seem to have improved, if anything it seems to have slightly increased.

⁴ Estimates of parental preferences generally found in the literature point to preferences being heterogeneous across socioeconomic groups with lower SES parents tending to put more weight on price and distance (Hastings et al., 2009; Gallego and Hernando, 2010).

⁵ In this paper, I abstract from neighborhood sorting. I focus on student demand and school selection as these are the mechanisms more likely to be affected by the 2008 educational reform used in this paper. Given the characteristics of the reform (see details in Section 2.1), it is unlikely to affect housing choices. Furthermore, Arteaga et al. (2018) show that in the Chilean context, residential segregation explains a very small part of overall school segregation.

Next, I explore schools' selective behavior in the admission process. Generally, although schools' selectivity based on socioeconomic characteristics is an issue openly discussed, evidence is limited due to the decentralized school application process (no data on individual applications and only enrollment is observed).⁶ The SEP program provides a useful opportunity to test for this cream-skimming behavior. To do so, I model schools' admission processes in terms of a threshold in admitted student's SES (I use maternal education as a proxy). Despite endogenous admission thresholds, the timing of the 2008 reform allows me to test for schools' selectivity. I analyze participating schools (as financing rules did not change for non-participating schools), comparing the type of students admitted before and after they join the program. I show that admission thresholds significantly decreased following the reform in SEP schools, even for those with no tuition prior to the SEP program for which there is no price change. This provides evidence that price changes alone cannot explain all the enrollment changes. Notably, this drop is only observed in schools likely to conduct parental interviews during the admission process. There is no change in the lowest type of student admitted for public schools or for private subsidized schools unlikely to conduct parental interviews.⁷ This sheds light on the mechanisms that schools have to extract information related to families' socioeconomic status. Consequently, low-SES parents who faced admission restrictions from private schools before the reform, had a larger set of choices post-reform. To the best of my knowledge, this is the first paper to provide evidence on school selectivity based on socioeconomic characteristics.

Estimating demand parameters is crucial for understanding drivers of student sorting and predicting allocation under alternative policies. I estimate a school choice model that incorporates admission restrictions at private schools based on student socioeconomic characteristics and allows for heterogeneous parental preferences for school characteristics and peer composition. This contrasts with previous work that assumes that parents can choose any school they are willing to travel to and pay for, which in practice attributes any sorting pattern observed in the data to demand-side preferences, rather than school selection.⁸ I show that ignoring admission restrictions underestimates low-SES parents' preferences for school quality. Furthermore, the changes in SEP schools' admission thresholds due to the new voucher create variation in peer composition, separately from quality characteristics. I use this variation to estimate parents' preferences and to study the effects of post-reform enrollment changes on segregation.

The estimates of the demand parameters show that low-SES parents value quality characteristics like test scores, class size, and peer quality. At the same time, high-SES parents have strong preferences for high-SES peers. A one standard deviation increase in peer quality gives ten

⁶ Most of the literature in the context of Chile that ignore school selection use the argument that in surveys few parents report being rejected from a school where they applied. This argument contradicts the extensive political discussions over the implementation of mechanisms to deter selection. In fact, a new law introduced in 2017 centralizes admission in order to prevent schools from selecting students at admission.

⁷ As schools admission processes are not observed, it is also unclear which schools conduct parental interviews. I use information from a survey to parents to infer which schools are more likely to request interviews during the admission process.

⁸ Hastings et al. (2009), Neilson (2021), and Gallego and Hernando (2010) also estimate parental preferences for school characteristics based on choices of schools, looking at heterogeneity in preferences across socio-economic groups. Several other papers estimate parental preferences for schools based on residential location (Black, 1999; Bayer et al., 2007). In the context of the United States, school selectivity is generally not a big concern given that public schools cannot discriminate students at admission and charter schools' admission is generally determined by a lottery. This is not the case in other countries like Chile where admission is decentralized and schools may have the ability to observe students demographic characteristics and admission decisions may be influenced by this information.

times as much utility to high-SES parents, as a one standard deviation increase in school average test scores. This is particularly interesting in light of the results shown in [Neilson \(2021\)](#) who studies the same reform. He shows that private subsidized SEP schools responded to the new voucher by increasing quality. This should, if anything, help retain high-SES students that have stronger preferences for school quality than low-SES students.⁹ Additionally, as mentioned before, for the subset of private subsidized schools that charge a top-up tuition, prices are increasing in non-SEP schools and decreasing in SEP schools, which should also work in the opposite direction.

These results map into the two different effects of the Chilean reform on student sorting that I described above when analyzing enrollment changes. First, it decreased admission thresholds at private subsidized SEP schools, which together with low-SES parents' preferences for better schools explain the increased enrollment of low-SES students in private subsidized SEP schools. Second, high-SES parents' preference for high-SES peers explain the increased enrollment in private subsidized non-SEP schools. These direct and indirect changes (of low-SES and high-SES students, respectively) followed distinct patterns. The first effect, caused by the financial incentives for private SEP schools, results in a discrete jump in the probability of low-SES students going to a SEP school immediately after the reform. The second effect, caused by high-SES parents responses to changes in peer characteristics, generated a gradual increase in the probability of high-SES students of enrolling in non-SEP schools. Overall, these changes prevented the intended decrease in segregation. If anything, there seems to be moderate increases in socioeconomic stratification in an already segmented educational system.

Finally, I use the model estimates to conduct counterfactual simulations to isolate the different mechanisms. My counterfactuals show that heterogeneous preferences for high-SES peers seem to be the main driver behind segregation. I show that eliminating cream-skimming by schools may further increase migration of students from public to private schools, with only moderate effects in segregation. The analysis focuses on demand responses and abstracts from modeling schools' pricing and admission threshold choices. The main reason to abstract from price changes is that a large majority of schools do not charge any tuition, and for schools that do have positive prices, there is a regulated maximum price schools are allowed to charge. On the other hand, given that the main counterfactual of interest is the elimination of school selectivity, there is no need to model cutoff decisions directly which are complicated to estimate in markets with several competitors. The main limitation of not modeling price responses is that my counterfactuals assume that prices do not change when school selectivity is eliminated. Nevertheless, given that I find a very limited effect of eliminating selection, allowing for price adjustments would only further attenuate this effect. The second counterfactual quantifies the importance of peer preferences. The point of this hypothetical exercise is precisely to isolate this response without allowing for equilibrium changes in school selection or prices. While expanding the model to study supply-side responses could be interesting for a broader set of counterfactuals and policy questions, it would add significant complexities and it is out of the scope of this paper.

This paper relates mainly to three strands of the literature. First, the literature on voucher education systems and in particular, the study of the relationship between school choice and socioeconomic stratification. While most research on vouchers focuses on student achievement, little is known about the stratification consequences of school voucher reforms and the mechanisms that explain socioeconomic sorting. In particular, how private schools respond to such policies in terms of admission criteria and which school characteristics drive parental choices.

⁹ This is a common finding in the literature. In the context of Chile, heterogeneity in preferences for academic quality is shown both in [Neilson \(2021\)](#) as well as in the estimates in this paper.

Studies on the SEP reform in Chile show improved test scores and a reduction in the achievement gap between high- and low-SES students ([Neilson, 2021](#); [Murnane et al., 2017](#); [Navarro-Palau, 2017](#)).¹⁰ These studies look exclusively at test-scores and abstract from the effects on student sorting. There is a small literature on school choice looking at the effects on segregation. [Söderström and Uusitalo \(2010\)](#) show that school choice in Sweden's secondary schools increased sorting by family background and segregation between immigrants and natives. [Hsieh and Urquiola \(2006\)](#) associate the implementation of the 1981 voucher reform in Chile with increased stratification across schools, but without being able to reveal the channels through which these effects may operate. Trying to make sense of the puzzle presented by the fact that more choice was not associated with achievement gains, they mention the possibility that parents choose schools based on peer characteristics, and schools may respond by selection processes along these same criteria. Using the variation in peer characteristics within schools induced by the SEP reform I am able to disentangle these mechanisms. My results align with the hypothesis that parental preferences for peer characteristics and selective school admissions drive sorting, helping explain persistent segregation in various public education systems across different countries.

Second, the results presented in this paper contribute to a large literature estimating parental preferences. Most studies highlight parents' heterogeneous preferences and the trade-offs between quality, distance, and tuition. High-SES parents often prioritize school quality, explaining their tendency to opt out of public schools under school choice programs ([Hastings et al., 2009](#); [Chakrabarti, 2013](#); [Böhlmark et al., 2016](#)). [Hastings et al. \(2009\)](#), [Chakrabarti \(2013\)](#) and [Böhlmark et al. \(2016\)](#). Yet, in general it is not clear what school quality means in each context encompassing both school and peer characteristics. [Abdulka-diroğlu et al. \(2020\)](#) estimate preferences for schools in New York City, and find that preferences are uncorrelated with school effectiveness measures when conditioning on peer quality. Consistent with my results, they find larger coefficients on peer quality for wealthier students. In these papers, variation in school quality and peer characteristics is entirely endogenous. The characteristics of the SEP program induce exogenous variation in student composition within schools, which allows me to explore further the preferences for peer quality among different types of students. The reform-induced reallocation of students generates within-school variation, useful for separately identifying preferences for school quality from taste for a given set of peers, a feature rarely observed.

Third, this paper also contributes to the literature on the implications of the design of school choice programs on sorting and stratification ([Hsieh and Urquiola, 2006](#); [Avery and Pathak, 2021](#); [Altonji et al., 2015](#)). [Avery and Pathak \(2021\)](#) and [Altonji et al. \(2015\)](#) study cream-skimming in school choice programs in the United States where, as mentioned above, there cannot be selection at admission from the schools' side. Therefore, all the sorting is explained by demand heterogeneity. Several papers on school choice have suggested deviating from a flat voucher design to alleviate cream-skimming and decrease segregation. [Nechyba \(2009\)](#) argues that efficient programs should incentivize competition through innovation and increased resource efficiency, rather than through selecting the best students from public schools. [Neal \(2002\)](#) and [González et al. \(2004\)](#) argue that vouchers that fall in value as household income rises may partially

¹⁰ The purpose of this paper is not to evaluate the SEP reform per se, which has shown mostly positive effects on student achievement (with the exception of [Feigenberg et al., 2019](#)), but to use the variation induced by the program to understand student sorting. Given the positive overall effects on test scores, the fact that it did not decrease student segregation may not seem of primary importance. Nevertheless, it is puzzling that decreasing the amount of selection from private subsidized schools did not improve student segregation. In general, understanding the mechanisms behind student sorting is key to improve the design of school choice policies.

offset incentives to cream-skin for competitive advantage. My results fill a gap in this literature by empirically looking at the sorting effects of targeted vouchers. The Chilean reform examined here effectively decreased cream-skimming, but unexpectedly had little effect on overall segregation. Though the reform sought to decrease inequality by giving more resources to schools serving low-SES students, it ignored the possibility of student re-sorting. Furthermore, given that most of the segregation is explained by heterogeneity of preferences, these results should attenuate concerns regarding school choice programs. Under regular neighborhood-based assignment, heterogeneity in parental preferences would likely result in higher rates of neighborhood sorting. Policy makers may have major challenges in reducing segregation if preferences for peer quality are so large in high-SES parents. This could be especially critical given evidence that school segregation perpetuates long-term income inequality (Benabou, 1996).

The remainder of the paper is organized as follows. Section 2 provides institutional background on Chile's education system and the 2008 reform. Section 3 describes the data. Section 4 shows a descriptive analysis of changes in school enrollment, segregation, and prices. Section 5 discusses schools' admission processes and provides evidence on schools' selectivity. Section 6 provides a general framework of the mechanisms behind socioeconomic segregation and describes the demand estimation in the presence of cream-skimming. Section 7 discusses the results, Section 8 shows some counterfactual exercises and Section 9 offers a summary and conclusions.

2. Background on Chile

Chile implemented a nationwide school voucher program in 1981 to introduce school choice and decentralize educational services. Under this program, students may choose among both public and private schools. Private schools not charging tuition began to receive the same per-student voucher from the government as did public schools. If a student decides to move to another school, the new school would receive the entire subsidy. Tuition-charging private schools continued to operate mostly without public funding, staying mainly unaffected by the reform. The share of students in private subsidized schools steadily increased from around 10% in 1980 to over 45% of all students in 2007. By that time, the share of students in private subsidized and in public schools were similar. The share of students in private non-subsidized tuition-charging schools has remained mostly unchanged at around 8%.¹¹

Starting in 1994 with the establishment of the *Financiamiento Compartido* program, private subsidized schools were allowed to charge a top-up in addition to the voucher with limits in the maximum amount they can charge in order to keep receiving the government subsidy. Still, by 2007, more than half of these schools do not charge anything. Appendix Figure A.1 shows the distribution of average tuition in private subsidized schools in 2007.

An extensive literature has studied the Chilean voucher program. Hsieh and Urquiola (2006) find no evidence that choice improved average educational outcomes as measured by test scores, repetition rates, and years of schooling and they show that the voucher program led to increased sorting. Contreras et al. (2010) offer some suggestive evidence that private subsidized schools are more selective than public schools. Even though there is extensive evidence of student sorting, it is less clear whether this is mainly explained by private subsidized schools' selective behavior or demand-side sorting. The empirical analysis of this paper shows evidence of both, schools selective behavior, but at the same time, strong heterogeneity in preferences from the demand side.

¹¹ See Bellei (2005), Patrinos and Quentin Barrera-Orsorio (2009), Elacqua et al. (2011) for details on how the system evolved the first 20 years.

Table 1

Increase in the value of the voucher for SEP students.

Source: Mineduc (2012).

	Preschool to 6th grade		
	2005–2007	2008–2011	2012
Baseline subsidy	\$68	\$79	\$81
SEP subsidy	\$0	\$46	\$56
% of priority students	Preschool to 6th grade		
15%–30%	\$3.6		
30%–45%	\$6.2		
45%–60%	\$8.3		
>60%	\$9.3		

Note: This table presents the values for the preferential subsidy for 2008 and 2011, and the extra voucher the schools get for a high concentration of priority students in US dollars per student per month.

2.1. The 2008 reform: SEP law

In response to critics of the old voucher system, in February 2008, Chile adopted a new policy creating a targeted schooling subsidy for the most vulnerable students (SEP law, for 'Subvención Escolar Preferencial'). The main objective of the SEP reform was to decrease education inequality (Mineduc, 2015). It modified the existing flat subsidy per student by introducing a two-tier voucher, with a higher subsidy for the most vulnerable students, trying to improve equity within the education system, promote equal opportunity, and improve the quality of education (Weinstein et al., 2010). Starting in 2008, participating schools received an extra voucher for students defined as priority by the SEP law. In addition, participating schools were required to design and implement an educational improvement plan. These schools were also required to accept the value of the voucher as full payment of tuition for preferential students, eliminating extra tuition and other fees for eligible students.

The monthly values of the extra subsidy are defined by the government and are adjusted for inflation every year, same as the original subsidy. These values are described in Table 1.¹² Student eligibility for the SEP voucher is determined annually by the Ministry of Education according to several criteria associated with socioeconomic vulnerability. By 2012, 44% of elementary school students were classified as eligible for the SEP benefits. Details about eligibility are explained in Appendix B.

Schools have the choice to register in the SEP program and only participating schools receive the SEP benefits. If a school chooses not to participate, it cannot receive the benefits even if priority students are enrolled, yet continues to receive the original voucher. SEP schools are required to adhere to several conditions, including the exemption to eligible students from any out-of-pocket expenses. This effectively decreased the price of participating schools that were previously charging tuition for eligible students. This is an important mechanism for student re-sorting that is discussed in the empirical analysis. In terms of participation, virtually all public schools and slightly over 65% of private subsidized schools registered in the SEP program. More details regarding the Chilean voucher program and the SEP reform may be found in Appendix B.

3. Data

The empirical analysis relies on administrative data on student enrollment together with data on school and student characteristics. I combine three data sources. The first is a comprehensive dataset on

¹² In addition, more resources were given to schools having a high concentration of priority students. This is also described in the second part of Table 1 which shows the resources assigned according to the concentration of SEP students in a school, on top of the baseline SEP subsidy.

Table 2
Number of schools by type and year.

	2005	2006	2007	2008	2009	2010	2011	2012
Total	7425	7457	7896	7764	7857	7890	7704	7612
Private non subsidized	409	399	406	406	401	406	405	393
Private subsidized	2694	2698	2923	2936	3060	3085	3094	3124
Municipal	4322	4360	4567	4422	4396	4399	4205	4095
All SEP	–	–	–	6553	6629	6649	6456	6328
SEP private subsidized	–	–	–	2137	2235	2252	2253	2237
SEP municipal	–	–	–	4416	4394	4397	4203	4091

Note: This table presents the number of schools each year by type of school starting in 2005, and the number of schools enrolled in the SEP program from 2008.

yearly school and student-level data from 2005 to 2012. It contains the universe of students and the schools where they are enrolled, along with some school characteristics. It reports the type of school, the concentration of SEP students in each school, which schools are registered in the SEP program, and the total amount of money received from the program each year. I exclude rural schools from the analysis because these schools face significant lower competition and are disproportionately public schools.

As mentioned above, school participation in the SEP program is voluntary. Table 2 shows the number of schools registered in the SEP program, together with the total number of schools per year. Almost all public schools, and over two thirds of private subsidized schools, participated in the SEP program after 2008.

Second, I use an additional dataset to construct other school characteristics like average test scores. This dataset contains SIMCE test results of all 4th grade students from 2005 to 2012. SIMCE is a standardized test taken by all 4th graders in the country.

Third, I use student demographic characteristics like family income, parental education, whether they have a computer and internet at home. This information comes from a questionnaire sent to the families of students taking the SIMCE test. This survey also includes a question regarding whether the parents had an interview at the school during the admission process. I use this information in the analysis of school admissions.

My analysis focuses on approximately 230,000 students per year enrolled in public and private subsidized schools.¹³ Table 3 shows descriptive statistics for student characteristics in private subsidized and public schools in 2007, before the SEP program started. Differences between students enrolled in the two types of school are apparent, students in private schools belong to wealthier families with more educated parents and on average obtain higher test scores (SIMCE scores have a mean of 250 and a standard deviation of 50). The table also reports descriptive statistics after the reform. By 2012, differences in parental education and family income were larger than differences in 2007, however the gap in test scores dropped significantly. These changes reflect student redistribution following the SEP reform, which are explored further below. Additionally, it suggests that the additional resources from the program may have had a positive effect on achievement.

3.1. Market definition

In the Chilean context, students are free to choose a school without geographic or administrative constraints, therefore there is no clear market definition. Distance is obviously a relevant variable, but how much students are willing to travel is heterogeneous and might depend on income, quality of public transportation, climatic conditions, etc.

¹³ I exclude private fee-paying schools from the analysis. These schools charge high tuition and do not receive any public funding, so they were mainly unaffected by the reform. They serve less than 8% of students, a share that did not change during the study period.

Table 3
Student characteristics for public and private subsidized schools in 2007 and 2012.

	Student characteristics in 2007			
	Public	Priv. Subsid.	Difference	T-Test
Mother's education (yrs)	10.29	12.47	–2.18	–34.24
Family income (US\$)	351.54	629.87	–278.33	–27.06
Math score	229.38	251.55	–22.17	–23.65
Language score	240.10	260.15	–20.05	–24.19
Class size	25.49	25.96	–0.47	–1.42
	Student characteristics in 2012			
	Public	Priv. Subsid.	Difference	T-Test
Mother's education (yrs)	10.06	12.15	–2.09	–37.91
Family income (US\$)	294.08	571.40	–277.32	–28.91
Math score	246.71	261.27	–14.56	–18.20
Language score	253.59	268.41	–14.82	–20.97
Class size	23.19	26.29	–3.1	–11.18

Note: This table presents average statistics in each type of school in 2007 and 2012. Column (3) shows the differences between these two years and the t-test for statistical significance. SIMCE test scores have a mean of 250 and standard deviation of 50. Averages are calculated over all students in 4th grade in the school.

I use data on student travel distance to define markets. For each school, I aggregate all municipalities where 5% or more of the students in that school live (restricting travel distance to a maximum of 200 kilometers).¹⁴ This creates a network of municipalities that constitute a market. There are a total of 28 non-overlapping markets under this definition. Some of these markets are individual cities with several municipalities, but in other cases, two cities located close enough are considered a single market.¹⁵

Appendix Figure A.2 shows a map of the central regions of Chile where most of the markets are located. The size of each bubble represents the market size in terms of number of schools and the table below the map shows the mean and standard deviation in the number of schools and students across markets, and the average distance traveled by students.

4. Descriptive analysis

In this section, I present descriptive evidence related to enrollment changes, measures of segregation, and price changes. I provide evidence of significant enrollment shifts following the implementation of the SEP reform. In particular, there was a decline in enrollment in public schools, an increase in enrollment of low-SES students in participating private subsidized schools, and a rise in enrollment of high-SES students in non-participating private subsidized schools. Furthermore, I show that contrary to the reform's objectives, these changes had little effect on overall segregation.

These findings raise the question about the mechanisms driving these enrollment shifts, particularly the increased enrollment in non-participating schools where voucher amounts remained unchanged. One simple explanation involves price responses: there might be higher prices for high-SES students in participating schools and/or a lower prices charged by non-SEP schools. However, the descriptive evidence indicates the opposite trend: prices are on average increasing in non-SEP and decreasing in SEP schools.

¹⁴ This is just a conservative limit to make sure that no relevant competitors are excluded.

¹⁵ I exclude small municipalities (with fewer than 10 schools) located far from other municipalities. Some small municipalities with few schools are included as long as they are close enough to other municipalities to constitute a market. The reason to exclude single-municipality markets is that the location of students is known only up to their municipality, where I assign the location of all students to the municipality centroid. Therefore, in a market with one municipality, all students would be located in the same point without meaningful variation to identify willingness to travel parameters.

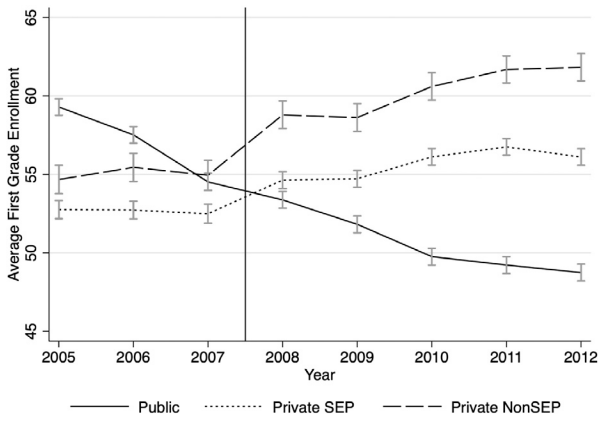


Fig. 1. Evolution of detrended first grade enrollment in public and private schools. Note: This figure shows the evolution of detrended first-grade enrollment within-school. It shows the coefficients on the year fixed-effects from a regression of first grade enrollment (detrended) in a school on year and school fixed effects. The constant from each regression is summed to the coefficients to facilitate the interpretation. The vertical line placed between 2007 and 2008 separates the years before and after the SEP reform.

In Sections 5 and 6.4, I analyze school-side selectivity and demand-side preferences, respectively.

4.1. Changes in enrollment

The SEP program sought to decrease educational inequality by giving more resources to schools serving vulnerable students. However, the changes introduced by the reform triggered significant re-sorting of students.

In this section, I distinguish between private subsidized schools that choose to participate in the program (private SEP schools) and private subsidized schools that choose to opt out of the program (private non-SEP schools). I show important student redistribution patterns following the 2008 reform between the three types of schools considered: public, private SEP and private non-SEP schools.

It is important to keep in mind that there were already some trends before the implementation of the reform. First, decreasing total number of primary school students in the country because of overall demographic trends. Second, enrollment decreasing faster in public schools than in private subsidized schools. Therefore, in order to analyze changes in enrollment associated with the SEP reform beyond demographic trends, I detrend school enrollment from country-level demographic changes.

Detrended first grade enrollment across different types of schools are presented in Fig. 1. It shows the coefficients on year fixed-effects from a regression of detrended first grade enrollment on school and year fixed effects (in the graph, I add the constant for each type of school for interpretation purposes), representing average within-school changes in enrollment:

$$\text{AverageEnrollment}_{jt} = \gamma_j + \eta_t + \varepsilon_{jt} \quad (1)$$

where γ_j and η_t are school fixed-effects and year fixed-effects, respectively.

The share of students in public schools steadily declines before and after the reform. In contrast, private subsidized schools increase their share of students around the time of the reform, both in SEP and non-SEP schools (average levels in 2005, 2006, and 2007 are almost unchanged and they increase significantly after 2008).¹⁶ The increased

enrollment in private subsidized SEP schools is expected. The new program creates incentives for these schools to admit more vulnerable students. Furthermore, the subset of private subsidized schools that had some tuition fees become free for SEP students. On the other hand, subsidies for private subsidized non-SEP schools are unchanged, they keep receiving the same original voucher for all students enrolled. Nevertheless, there are several factors that could explain an increase in enrollment in non-participating schools like changes in prices, other school characteristics, or peer composition. Disentangling demand preferences is the focus of the model in Section 6.4.

Additionally, changes in enrollment are not homogeneous across types of students and they occur at different times. Fig. 2 shows the probability of going to each type of school for different student types across the sample period. These probabilities are calculated from a multinomial logistic regression where the choice options (K) are the four types of school (public, private SEP, private non-SEP, private fee-paying) and the explanatory variables are mother's education and year fixed-effects:

$$\text{Prob}(it \rightarrow K) = \frac{\exp(\beta_K \theta_i + \eta_{Kt})}{1 + \sum_{k=2}^4 \exp(\beta_k \theta_i + \eta_{kt})}$$

where $\text{Prob}(it \rightarrow K)$ is the probability of student i attending a school of type K .

In this very simple specification, mother's education enters linearly. Appendix Figure A.5 adds a squared-term which only has some noticeable effect at the extremes for very low or very high types.

Each line in the graphs corresponds to a different year, with solid lines showing years before the reform and dashed lines the years after the reform. To facilitate the interpretation, three vertical lines mark the 25th, 50th, and 75th percentile in the type distribution (the median is 12 years of mother's education).

On the year the reform was implemented, the probability of going to a public school dropped significantly for students in the bottom half of the distribution of mother's education (upper left graph), with a corresponding rise of similar magnitude in the probability of going to a private subsidized SEP school (upper right graph). There is an increase of about 10 percentage points in the probability of going to a private SEP school for students with mothers with fewer than twelve years of education. This change occurs in a discrete way, starting in 2008, the first year of the SEP reform, which suggests that private SEP schools started admitting students they had not admitted before. Furthermore, it suggests that the high enrollment in public schools of students in the bottom half of the distribution before the SEP reform, was likely determined by their inability to meet private schools admission requirements (either through direct selection at admission or tuition in schools that were charging a fee).¹⁷

Additionally, there is a gradual increase in the probability of students in the middle-high part of the distribution of mother's education (above 75th percentile) of going to private non-SEP schools in the years following the implementation of the reform (bottom left graph). This contrasts with the sharp rise in the probability of enrollment in private SEP schools for low-SES students. This gradual rise in probability of going to private subsidized non-SEP schools for students with more educated parents is likely associated with changes in the characteristics

some outliers, I also run the same regression using log of enrollment as the dependent variable with very similar results (see Appendix Figure A.4).

¹⁷ An alternative but similar explanation relates to participating schools actively pursuing low-SES students because of the new subsidy, as opposed to selecting them out before 2008. I discuss this in Section 5 where I provide some evidence against this argument. In any case, for the demand estimation these two explanations have analogous implications, lack of information about schools available or being selected out would in practice mean that some schools are not being considered in their choice set by low-SES students. Therefore, for identification of the model parameters, the distinction between them is not of primary relevance.

¹⁶ There is significant heterogeneity in school size, both among public and private subsidized schools (see the distribution of first grade enrollment in Appendix Figure A.3). To assure that the results in Fig. 1 are not driven by

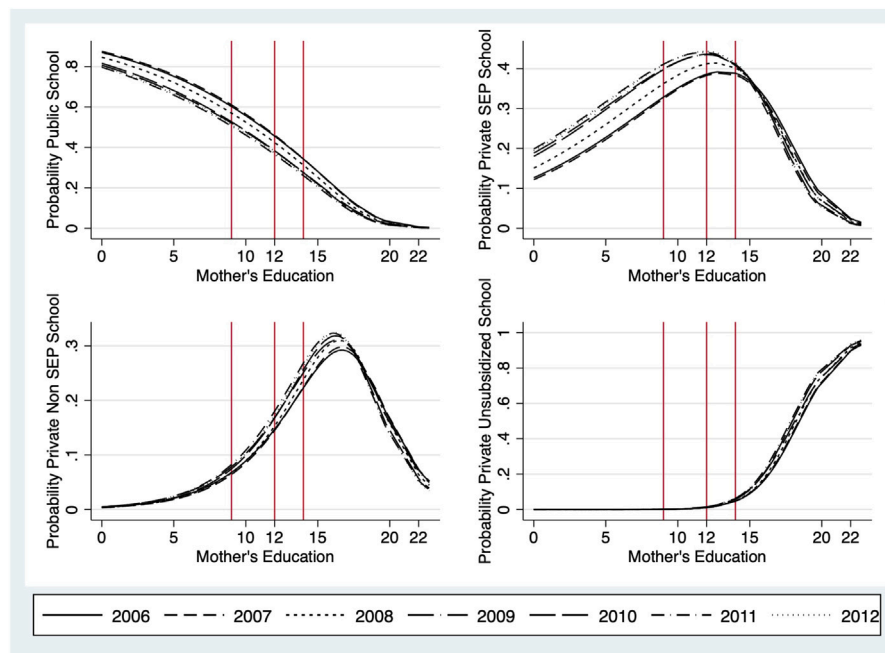


Fig. 2. Probabilities of enrollment by type of school. Note: Each graph shows the probabilities for each level of mother's education of enrolling in different type of school each year. The probabilities are calculated based on the coefficients from a multinomial logit model where a student has the option of enrolling in four types of schools: public, private subsidized SEP, private subsidized non-SEP, and private fee-paying schools. For each type of student-year, the probabilities across the four types of schools sum to one. The three red vertical lines mark the 25th, 50th, and 75th percentile in the distribution of student types.

of SEP schools following the reform. If highly educated parents have preferences for certain types of peers, changes in admissions requirements by private SEP schools, admitting more vulnerable students, may have led high-SES parents to stop choosing private SEP schools and enroll instead in private subsidized non-SEP schools.

For completeness, the bottom-right graph shows the probability of going to a private unsubsidized school. There is a small gradual increase in these probabilities, mostly explained by a very small percentage of students at the top of the distribution (less than 7% of students attend this type of school, and less than 1% of students have a mother with more than 17 years of education).

A possible concern with these descriptives relates to capacity constraints, as schools are unlikely able to increase significantly their capacity in the short term it is puzzling that the probabilities of low-SES to enroll in SEP schools could increase so much. One important issue to keep in mind are the strong decrease in cohort size during this time period. Cohort size has decreased by 15% from 2004 to 2012, therefore most schools have excess capacity and can increase their shares without significantly increasing their capacity. Some schools in the data do add additional classrooms, but these are not the majority.

4.2. Changes in tuition

A possible explanation for the increase in enrollment in non-SEP schools could be related to price changes. SEP-schools could increase top-up tuition to high-SES students to compensate for the voucher difference between low- and high-SES students. On the other hand, non-SEP schools could also respond by changing their prices. If SEP schools were increasing prices for high-SES students, and/or non-SEP schools were decreasing prices, these could explain the increase in enrollment in non-SEP schools.

I run the same regression from Eq. (1) using prices as the dependent variable on school and year fixed effects. Fig. 3 shows the coefficients from this regression (in the graph, I add the constant for interpretation purposes), representing average within-school changes in tuition. It shows exactly the opposite. On average, prices are decreasing in SEP schools and increasing in non-SEP schools. This is consistent with

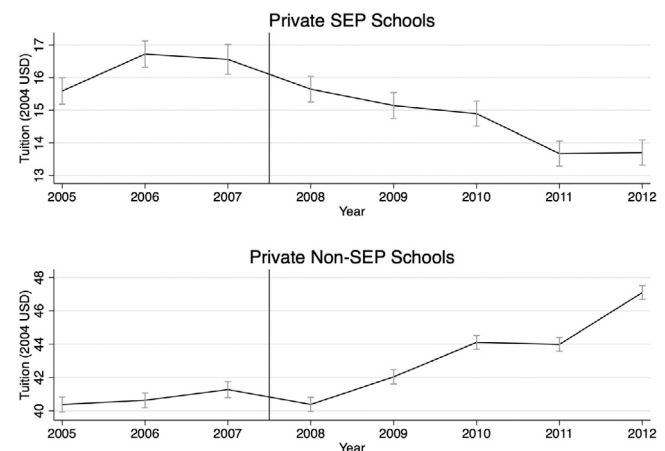


Fig. 3. Top-Up fees. Note: This figure shows the evolution of top-up fees within-school. It shows the coefficients on the year fixed-effects from a regression of first grade enrollment in a school on year and school fixed effects. Fees are expressed in 2004 USD using a constant exchange rate. The constant from each regression is summed to the coefficients to facilitate the interpretation. The vertical line placed between 2007 and 2008 separates the years before and after the SEP reform.

demand increasing for non-SEP schools. In the figure, it is also apparent the differences in price between participating and non-participating schools.

4.3. Segregation measures

In order to quantify changes in socioeconomic segregation resulting from student sorting during the years around the SEP reform, I compute several measures commonly used in the literature. I use two types of measures to capture different aspects of stratification. First, I compute segregation measures at the market level by school type. This involves examining changes in the average student type in public schools compared to the average student type in the overall market. In the second

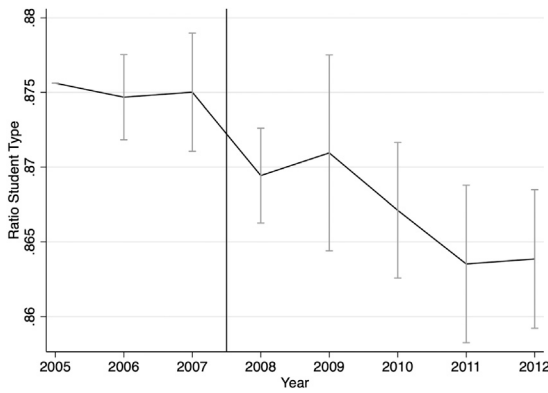


Fig. 4. Changes in average SES in public schools compared to market average. Note: This figure shows average changes in the sorting ratio measure. The ratio is constructed as the average student type in public schools over the average student type in the market where the public schools operate. Higher values mean more integrated markets where public schools have a more representative student body compared to the market where they operate. Each point in the graph represents the average ratio across markets, weighted by total number of students in the market. The vertical line placed between 2007 and 2008 separates the years before and after the SEP reform.

part of this section, I compute segregation measures at the school level using the dispersion of students within schools and the dissimilarity index (in Appendix D). None of these measures indicate any positive effect on reducing stratification. If anything, the evidence presented here suggests small increases in segregation across schools.

The reason for first examining market-level measures is due to the common concern that Chile's school choice system allows private subsidized schools to cream-skim students with higher socioeconomic characteristics. Hsieh and Urquiola (2006) show that the average parental education in public schools is significantly lower than in private subsidized schools, and this disparity is more pronounced in municipalities with higher private school enrollment. Using this same measure, I compare the average mother's education in public schools to the average mother's education in the overall market to test for each market-year pair to determine whether sorting changed after the SEP reform was implemented in 2008. Values closer to one indicate more integrated markets, whereas lower values indicate more segregated markets, where public schools serve a lower socioeconomic population compared to the market average. Note that this measure is not bounded by one. If public schools had the most educated parents in the market, the measure would exceed one, indicating segregation in the opposite direction. The SEP reform intended to improve this measure by reducing the incentives for private subsidized schools to select out students from low-SES backgrounds.

Fig. 4 shows the average ratio weighted by market size from 2005 to 2012, with confidence intervals for the difference relative to 2005. The average socioeconomic status of students in public schools compared to the market average decreased sharply in 2008 and continued to decline thereafter. This indicates that the increased enrollment in private subsidized schools was not due to a random sample of students who would have otherwise attended public schools, but rather to students with higher socioeconomic characteristics, further increasing the disparity in student types across different schools types. To interpret these numbers in terms of parental education, the average socioeconomic status of students in public schools decreased by 0.2 years of detrended mother's education. This number varies by market, with a maximum drop of 0.7 years, where the highest declines are observed in large markets. In 21 out of 28 markets, the average type of student in public schools declined. This is consistent with the admission model in Section 6 where private subsidized schools have a threshold for admission that decreased following the SEP reform in participating schools.

To measure school segregation at the school level, I analyze the dispersion of student types within schools. If markets become more

stratified, we would expect schools to become more homogeneous. For each school-year, I compute the interquartile range (IQR) of student types, defined as the difference between the 25th and 75th percentile of mother's education in each school year. I then run the following regression to capture changes within schools for each of the three types of schools: public, private subsidized participating in the SEP program, and private subsidized opting out from the program.

$$IQR_{jt} = \gamma_j + \eta_t + \varepsilon_{jt}$$

Fig. 5 shows the coefficients on the year fixed effects for each of the three regressions. These coefficients can be interpreted as average within-school changes in IQR compared to 2005 (the omitted category for the year fixed effects). Consistent with the enrollment changes discussed above, public and private SEP schools show a decrease in student type dispersion within schools, while no significant change is observed for private subsidized non-SEP schools. In terms of magnitudes, the baseline IQR is 4 for public schools, 3.5 for private SEP schools, and 3.2 for private non-SEP schools. Public schools experienced a decrease of 0.15 years from 2009 onwards, equivalent to 10% of a standard deviation of IQR across schools. For private SEP schools, the larger drop of 0.3 occurs from 2010 onwards. Interestingly, there are no significant changes for private non-SEP schools, suggesting that even with increased enrollment, the type of student attending these schools is not changing significantly.

The most relevant comparisons in this figure are between neighboring coefficients (rather than comparisons with respect to zero). For public schools, the difference between 2008 and 2009 is statistically significant (as well as between 2008 and the following years). On the other hand, for private SEP schools, there is a significant difference between 2009 and the three subsequent years. This timing aligns with the evidence shown in Fig. 2. For public schools, the most significant changes occur the year after the SEP reform implementation, while for private SEP schools, larger changes are observed starting in 2010. In summary, public and private SEP schools appear to become more homogeneous after 2009–2010 consistent with the changes from Fig. 2.

Additionally, Appendix D shows other segregation measures using the dissimilarity index to quantify the evenness with which low-SES students are distributed across schools (Duncan and Duncan, 1955; Reardon and Firebaugh, 2002). The findings remain robust across the different measures.

Overall, there is no evidence of any decrease in segregation. Examining a diverse set of measures at both the market level and individual schools level, it seems clear that the reform did not achieve its intended effect towards reducing stratification across schools. The evidence presented here points to small increases in segregation, primarily due to the lowest SES students remaining in public schools, mid-low SES students gaining access to private subsidized SEP schools, and higher SES students increasingly enrolling in private subsidized non-SEP schools.

Several mechanisms could explain these changes. On the supply side, there are price drops in SEP schools for eligible students, while prices in non-SEP schools appear to be increasing. Beyond price changes, SEP schools may have altered their admission policies in response to the additional voucher from the SEP the program. Changes in school admissions are analyzed in Section 5. Additionally, changes in school characteristics or peer composition could influence parents' choices. If parents have preferences for certain peer types, this could explain the increased enrollment in non-participating schools.

To understand what drives enrollment decisions, a demand model is needed to study preferences for different school characteristics and peer quality. Moreover, if the changes in enrollment for low-SES parents shown in the descriptive analysis are not entirely explained by price changes but also by changes in private school selectivity, it is important to account for these restrictions in low-SES parents' choice sets to correctly estimate parental preferences. In Section 6, I present a demand model that incorporates restrictions in the choice sets due to supply-side selection.

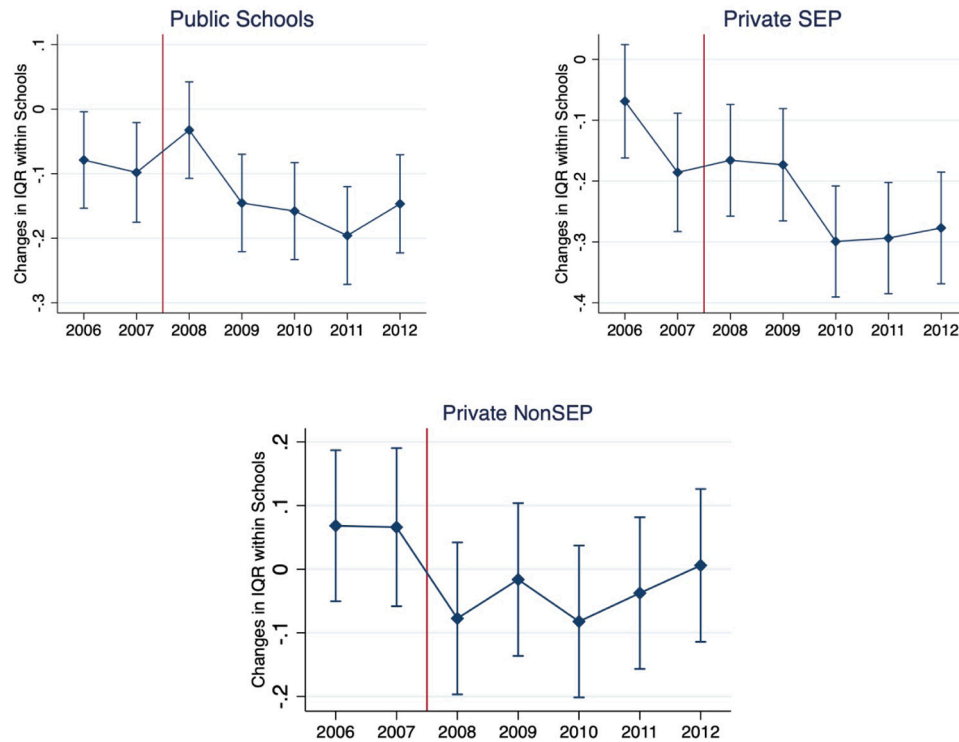


Fig. 5. Changes in within-school interquartile range of student SES. Note: This figure shows average change in the interquartile range (IQR) of mother's education within school for public, private SEP and private non-SEP schools. Each graph represents a separate regression and each point in the graphs shows the year fixed-effects coefficients from a regression of IQR in a school on year and school fixed effects. The vertical line placed between 2007 and 2008 separates the years before and after the SEP reform.

5. School admissions

In this section, I explore evidence of schools' selective behavior during the admission process. Correlations between academic performance and socioeconomic characteristics create incentives for schools to select students from more educated families. This selective behavior is also consistent with extensive political discussions about implementing mechanisms to deter selection and the observed stratification shown in the descriptive analysis. Nevertheless, the literature that estimates parents' preferences for school quality in Chile assumes that schools' only form of selection is through pricing. Yet, most of these schools do not charge any tuition.¹⁸

Schools' selective behavior is generally difficult to demonstrate, mainly because the admission process is decentralized and there is no data on individual applications (only enrollment is observed). Therefore, admission criteria remain unobserved. The admission process is highly opaque, and it is unknown whether sociodemographic characteristics are observed by the schools, and if they are, whether this information influences admission decisions. Furthermore, there is significant heterogeneity in the lowest type of students enrolled in each school within public and private subsidized schools. See Figure C.1 in Appendix C for a distribution of lowest types before the implementation of the reform.

In this context, the SEP program provides useful variation in the incentives for participating schools to admit more vulnerable students. The question is whether schools have the means to react given that they do not have measures of student performance at the moment of admission (for elementary school students). Furthermore, by law, elementary schools are not allowed to test students during the admission process. If schools are selecting students, discrimination has to be based on other

indicators, one of the easiest being parental education, likely observed through parental interviews.

I exploit the variation in incentives for schools that participate in the SEP program to test for whether schools are able to respond by lowering their admissions thresholds. Using mother's education as a proxy for the student type, I show that there is a significant decrease in the lowest type of students being admitted to participating private schools following the reform. This indicates that these schools are, in fact, admitting more low-SES students after the reform, consistent with a drop in admission thresholds. Yet, part of this drop may be explained by a price effect, given that the new program prevents schools from charging any tuition to eligible students.¹⁹

In Eq. (2), I define θ_{jt}^* as the admissions threshold, which in reality is unobserved. I use the first percentile of student types in each school-year (types defined by mother's education), as the 'observed' θ_{jt}^* and examine how it changes among participating schools when they join the program (schools join in different years, not necessarily all in 2008). I estimate Eq. (2) by OLS using school fixed-effects and $\mathbb{1}(SEP-School)_{jt}$, an indicator for whether school j is participating in the SEP program in year t . To exclude changes that may be explained by a price drop, I also restrict the sample to only schools that were not charging tuition before the reform was implemented.

Since many schools have fewer than 100 students per cohort, the first percentile often coincides with the minimum, making it a noisy statistic. To test the robustness of these results, I also use the second-to-last, and third-to-last values of mother's education as the dependent variable showing very similar findings.

$$\theta_{jt}^* = \alpha + \beta \mathbb{1}(SEP-School)_{jt} + \gamma_j + \varepsilon_{jt} \quad (2)$$

¹⁸ Approximately half of the private subsidized schools do not charge any tuition, and 75% charge less than \$30 per month; see Figure A.1.

¹⁹ Private schools participating in the SEP program need to exempt eligible students from all fees, including any tuition they charge to other students. For more details, see Section 2.1.

Panel A in Table 4 shows the average within-school change in the observed first percentile of mothers' education for SEP schools when they start participating in the SEP program. The first and second columns present the results for public schools and private subsidized schools, respectively. There is a large drop in the lowest type of students enrolled in private schools after they join the reform, but not in public schools. Column 3 shows the results using the restricted sample of schools that were not charging tuition to any student before the reform. Even among tuition-free schools, there are significant changes in the lowest parental education of admitted students, indicating that not all enrollment changes can be explained by a price effect.²⁰

Given the heterogeneity among voucher schools, it is unlikely that all of them use these admission practices. As schools likely obtain information regarding applicants' socioeconomic status via parental interviews, this effect should be more evident in schools that conduct interviews. Although this is unobserved, a survey to parents conducted alongside the SIMCE test asks whether they had an interview during the admission process. Among students in voucher schools, 37% of parents reported having an interview. Within a school, not all parents respond the same (only about 2% of schools have all parents responding uniformly to this question). I use this information to classify voucher schools into two groups: "likely to perform interviews" and "unlikely to perform interviews", based on whether more than 50% (or 40%) or less than 15% (or 25%) of parents, on average, responded affirmatively to this question. Columns 4 to 7 in Table 4 show the estimates for the same regression for each of these groups of schools (again restricting the sample to include only schools that never charged tuition). Interestingly, only the schools likely to conduct parental interviews during the admission process show a significant drop in the lowest type of students admitted after joining the SEP program.

As a robustness check, Panels B and C of Table 4 present similar results using the second-to-last and third-to-last student type, respectively, with very similar findings. Public schools show a larger change in these cases compared to panel A, indicating an increase in the lowest types, which is consistent with some low-SES students enrolling in private SEP schools.

These results provide evidence that schools are effectively able to select students based on socioeconomic characteristics likely observed during parental interviews. They also suggest that mother's education can serve as a useful proxy for the socioeconomic variables relevant to schools' selection processes. Table C.2 in Appendix C provides further evidence that this is a good proxy for how schools set admission thresholds. I show that only over-demanded schools exhibit a significant drop in θ_{jt}^* , consistent with school selection, as these are the schools more likely to select students. Although, I do not observe school capacity directly, I infer it based on historical enrollment. More details on this additional analysis and robustness regarding the definition of socioeconomic status are provided in Appendix C.

To further explore the timing of the changes in admission, Fig. 6 shows average predicted values for the first percentile of mother's education, estimated from a regression of θ_{jt}^* on school fixed-effect and year fixed-effects. The figure on the left shows the evolution of cutoffs for public schools, where there is no clear trend. The figure in the middle shows the same for private subsidized SEP schools, and the figure on the right shows the trend for the subset of private SEP schools that did not charge any fees before the implementation of the program. In both figures, there is a clear downward trend starting in 2008.

The last figure shows that admission thresholds, on average, drop in the first year after the implementation of the SEP program and remain low in subsequent years. Since these schools were not charging any fees even before 2008, this effect cannot be explained by changes in

tuition. Appendix Figure C.2 shows the same graphs using the second-to-last student type in each school instead of the first percentile, which indicates a similar trend.²¹

6. Demand model

This framework aims to capture the main mechanisms that could drive the sorting patterns shown in Section 4. In this model, all parents choose an elementary school for their children in a setting where public and private schools compete for students. The focus of the model is on the demand side to capture the heterogeneity in preferences. The evidence from Section 5 shows that schools can select students based on socioeconomic characteristics, which means that not all students have the same set of schools to choose from. Therefore, I assume that private schools are able to select applicants at admission based on observable characteristics, resulting in different choice sets for students of different socioeconomic statuses.

6.1. Overview of the model

A market consists of a set of students I and a set of schools J , where each student i attends a school j . Students are characterized by their type θ , which is determined by their socioeconomic status. The application process to schools is decentralized, and parents need to apply to each school they are considering through a separate admission process. Public schools are assumed to admit all students that apply, but private schools have the capacity to select students. Since these are primary schools, they do not observe students' abilities but do observe families' sociodemographic characteristics. Most schools are tuition-free and financed through government per-student vouchers paid directly to each school (both to public and private subsidized schools). I model parents' decisions as a discrete choice of a single school from their market.

6.2. Demand

The utility student i gets from attending school j at time t is denoted by U_{ijt} , which is modeled as:

$$U_{ijt} = \alpha_i \cdot p_{ijt} + \gamma_i \cdot d_{ij} + \mathbf{X}_{j,t-1} \beta_i + \lambda_i (\theta_i - \bar{\theta}_{j,t-1}) + \xi_{jt} + \varepsilon_{ijt}$$

where d_{ij} is the distance for student i to school j and p_{ijt} is the top-up tuition that the school charges. Importantly, tuition differs within a school for different students because of the SEP requirement to exempt eligible students from any charges. Therefore, for participating schools, tuition is set to zero for the SEP eligible students.

$\mathbf{X}_{j,t-1}$ are lagged observed school characteristics, which includes: test scores, class size, average peer type in the school, and the interquartile range of peer types in the school (as a measure of student heterogeneity). Additionally, $(\theta_i - \bar{\theta}_{j,t-1})$, the difference between the student's own type and the average type in the school, is included to allow for relative preferences for certain type of peers. For all the observable school characteristics, I use lagged values because this is the information available to parents when making schooling decisions, abstracting from any social interactions that may affect the decision.

²¹ In this paper I abstract from studying schools' participating decisions in the SEP program. This is an endogenous decision likely influenced not only by schools' characteristics but also by competitors' decisions. For this part of the analysis, I focus solely on changes within the group of participating schools, for whom the changes in financial incentives are significant. Therefore, the endogeneity of participation should not affect the interpretation of these changes in admission criteria. It is important to note that non-participating schools are likely different and are treated as such throughout the paper. In Appendix E, I explore the factors that may influence participation in a descriptive way. Modeling the schools' decision in a rigorous way is beyond the scope of this paper.

²⁰ Results are similar if I use a measure of socioeconomic status constructed using factorial analysis taking into account family income and education of both parents. See Table C.1 in Appendix C.

Table 4
Changes in admission thresholds.

	Public Schools	Private subsidized SEP schools		% of parents in the school reporting admission interview & Tuition = 0			
		All	Tuition = 0	>50%	>40%	<25%	<15%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A Dep variable: Lowest 1% of student mother's education							
SEP school	0.073* (0.034)	−0.413*** (0.043)	−0.301*** (0.063)	−0.412*** (0.121)	−0.385*** (0.108)	−0.124 (0.101)	0.003 (0.151)
Constant	3.169*** (0.027)	5.265*** (0.030)	4.125*** (0.046)	4.550*** (0.084)	4.489*** (0.076)	3.709*** (0.071)	3.699*** (0.103)
N	13,204	12,082	4742	1491	1946	1867	897
Panel B Dep variable: Second to last of student mother's education							
SEP school	0.244*** (0.030)	−0.310*** (0.034)	−0.181*** (0.052)	−0.293** (0.099)	−0.333*** (0.089)	−0.042 (0.087)	−0.109 (0.129)
Constant	4.563*** (0.024)	6.867*** (0.024)	5.677*** (0.038)	6.065*** (0.068)	6.038*** (0.062)	5.331*** (0.061)	5.464*** (0.088)
N	13,204	12,082	4742	1491	1946	1867	897
Panel C Dep variable: Third to last of student mother's education							
SEP school	0.338*** (0.027)	−0.283*** (0.030)	−0.168*** (0.048)	−0.150 (0.086)	−0.242** (0.079)	−0.046 (0.081)	−0.078 (0.129)
Constant	5.488*** (0.022)	7.841*** (0.021)	6.701*** (0.035)	6.994*** (0.060)	6.990*** (0.055)	6.420*** (0.058)	6.518*** (0.088)
N	13,204	12,082	4742	1491	1946	1867	897

Note: This table presents changes in the observed lowest 1% in the mother's education in Panel A (second to last and third to last in panels B and C, respectively) when schools join the SEP program. The estimation is based on a regression with school fixed effects to capture variation within schools, showing that mother's education is a good proxy for the admissions threshold. Column (3) uses the sample of private subsidized schools that never charged tuition. Columns (4) to (7) classify private subsidized schools depending on the percentage of parents that reported having an interview during the admission process. This information comes from a survey administered to all families conducted along with the SIMCE test. Columns (4) and (5) include schools likely to have parental interviews as more than 50% and 40% of parents, respectively, report having an interview during the admission process. Columns (6) and (7) include schools unlikely to conduct interviews as less than 25% and 15% of parents, respectively, report having an interview. *, **, and *** denotes significance at the 10, 5 and 1% level. Standard errors are presented in parenthesis.

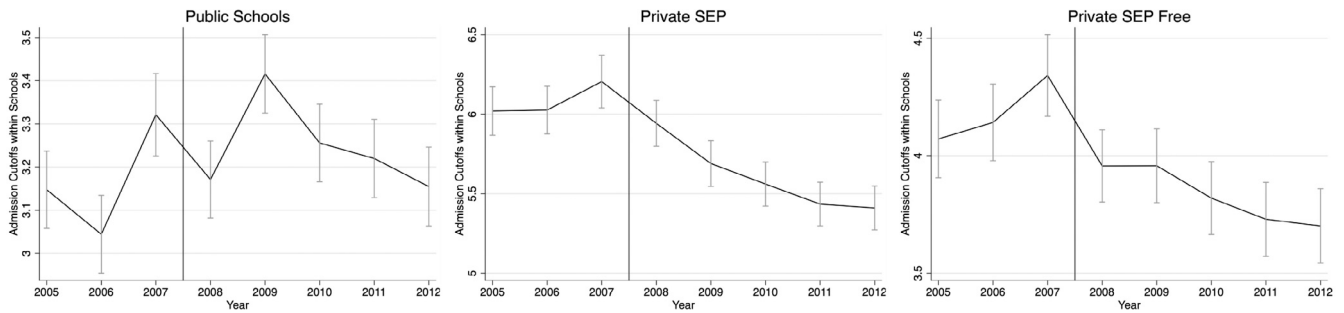


Fig. 6. Changes in admission thresholds. Note: This figure shows average changes in the first percentile of maternal education of admitted students for 2005 to 2012. Each graph represents a separate regression of the first percentile of mothers' education in a school on year and school fixed effects. Each point shows predicted values and standard errors. The graph on the left shows public schools, the graph in the middle shows all private participating schools, and the graph on the right shows the subset of private participating schools that did not have any fees before the implementation of the program. The vertical line placed between 2007 and 2008 separates the years before and after the SEP reform.

This is supported by the descriptive analysis shown in Fig. 2, where changes in enrollment probabilities for middle- and high-SES families seem to gradually respond to changes in school characteristics that followed the SEP reform. Additionally, using lagged characteristics is useful for identification purposes, as will be discussed further below. Similar reasoning supports the use of average test scores (as opposed to more sophisticated measures of school effectiveness) on grounds that this is the information more widely available to parents when applying to schools.

Unobserved school quality ξ_{jt} is decomposed in two terms, time invariant ξ_j and year-school specific idiosyncratic shock $\Delta\xi_{jt}$. ε_{ijt} represents an unobserved idiosyncratic preference of student i for school j , distributed independently across schools and students.

Parents are allowed to have heterogeneous preferences according to observable characteristics (income and parental education). This is reflected in parameters α, β, γ , and λ indexed by i , where,

$$\alpha_i = \alpha^o W_i$$

$$\beta_i = \tilde{\beta} + \beta^o W_i$$

$$\gamma_i = \gamma^o W_i$$

$$\lambda_i = \lambda^o W_i$$

W_i includes family income and mother's education. Income is a continuous measure of per-capita family income, and mother's education include indicators for being in one of four groups: less than eight years, less than high-school, high-school or more, and university degree. The omitted category is less than eight years.

Parents choose the school that maximizes their utility among the set of schools available to them given their type. Availability is determined by schools' admission policies discussed in the next subsection.

6.3. Private school admissions

Private school admissions are modeled as cutoffs on the socioeconomic type of students that individual schools are willing to admit. For

the demand estimation, these cutoffs are assumed to be observed in the data and allowed to adjust flexibly in the counterfactual analysis. In this subsection, I provide a simple model that rationalizes the existence of these admission thresholds.

Under a voucher program, public and private subsidized schools compete in each market to attract students.²² All schools (both public and private subsidized) receive a voucher v from the government per student enrolled. Each school has an unobserved capacity q_{ji} . Given a set of applicants, schools need to decide who to admit to fill up that capacity. Schools' utility depends on the number of students (which determine the funds received from the government), and the type of students registered. It is assumed that elementary schools cannot directly observe achievement during the admission process (because students do not yet have an academic record), but they are able to observe applicants' types. Evidence of this selectivity is presented in Section 5

There are two main mechanisms linking students' types and schools' utility. First, student achievement is known to be positively correlated with socioeconomic status, with higher types performing on average better than lower types. Therefore, the marginal cost of educating a student is decreasing in the type.²³ Second, if parents have preference for schools with higher test scores and/or for peers of higher types, demand for a school will be affected by the decision of which types to admit. Given these mechanisms, schools would prefer enrolling higher types to lower types.

Therefore, private subsidized schools will give priority in admissions to higher types over lower types among their applicants. Schools' capacity is not fully flexible in the short term, which translates into a cutoff rule in the socioeconomic type for admission. Conditional on the school's capacity, admission thresholds adjust endogenously to clear the market, working like prices in a regular supply and demand model.

Modeling schools' admission policies as a threshold on student socioeconomic type has the advantage of capturing schools' selectivity in a simple and flexible way. Furthermore, this is supported by the empirical evidence shown in Section 5 based on participating schools' responses to the new policy. I do not directly model the schools' optimization problem to define the capacities.²⁴ This is a complicated game with several differentiated players in each market (even a small market has at least 20 private subsidized schools horizontally differentiated competing with each other in an oligopolistic way). When schools are making their decisions regarding the type of students to admit, they have to consider the impact over marginal costs and over students' demand.

This is arguably a simplified model of the reality behind the admissions process, but it is able to capture schools' selectivity in a flexible way. There are some examples in the literature of estimation of admission probabilities in a decentralized setting with multiple schools competing against each other via tuition and admission policies (Fu, 2014). In that setting, in order to keep the estimation feasible, colleges are aggregated into four groups and admissions policies are defined as probabilities of admission for different types of students to each type of school. This has the cost of assuming that colleges within a type are identical (except for location). In the Chilean elementary school setting, schools are small and very heterogeneous, so it would not be straightforward to group them in a few different types. Estimating

school specific admission probabilities for different types of students would be unfeasible without making strong simplifying assumptions. The admission process modeled here is flexible enough to allow for schools reacting heterogeneously to the new policy and to changes in demand.

For estimation purposes, the assumption I make translates into including in a student's choice set all schools where at least one student of the same type or lower type is enrolled. Appendix Figure A.6 shows the average share of schools available for a student depending on their type. Students whose mothers that have at least finished high-school would have practically all schools available, whereas for students whose mothers have 8 years of schooling, the share of schools available is around 60%.

6.4. Demand estimation

The assumption commonly made in the literature estimating parental preferences is that every school is available to every student. Nevertheless, the evidence shown in Section 5 suggest that choice sets of feasible schools are different for low-SES than for high-SES parents.²⁵ Even if some schools are tuition-free, they might not admit some students based on their socioeconomic characteristics. Therefore, assuming that all students have the same choice set is not innocuous and it biases the estimation of preferences (for example, if high-quality schools are more selective, low-SES preferences for quality would be downward biased).

It is necessary to define the set of feasible schools for each student. If admission thresholds were observed, the choice set for each student would be easily defined. In this context, admission criteria in each school is unknown and could differ across schools. As explained in Section 5, most private subsidized schools have interviews for applicants' parents and it is reasonable to assume that during these interviews, schools are able to observe some socioeconomic characteristics. In that same section, I show that mother's education is a good proxy for characteristics that are relevant for admission to private subsidized schools, and that the lowest type behaves as a threshold under several different scenarios. Therefore, for the estimation, I make the assumption that the admission threshold for a school θ_{ji}^* is observed, equivalent to the minimum mother's education of enrolled students for each private subsidized school in a given year. In this context, admission thresholds are endogenous equilibrium outcomes, but this should not affect the estimation of parental preferences. Fack et al. (2019) show that estimates do not change when endogenizing the cutoffs, suggesting that in large enough markets, cutoffs can be treated as exogenous to estimate the demand parameters. Therefore, if equilibrium thresholds were known, estimates of utility parameters could be obtained by restricting the choice set for each student to the feasible schools only.²⁶ Importantly, once recovered the utility parameters, for the counterfactual exercises, admission thresholds are allowed to endogenously adjust.²⁷

Clearly, not observing the feasible choice set of schools for each student makes the demand estimation more challenging. The approach

²² Markets are defined geographically, more details on market definition are given in Section 3.1.

²³ Additionally, higher parental education is associated with better student behavior, more involved parents, the ability to attract better teachers, higher test scores, etc.

²⁴ Private schools are heterogeneous entities, therefore their utility functions are likely to differ across different schools. Some schools are profit-maximizers, but many schools are non-profit institutions. Regardless, even for non-profit organizations, schools care about rankings that are based on results from nationwide standardized test scores.

²⁵ The average choice set size (market-weighted average number of private subsidized schools available) for each type of student is 10 for the lowest SES category (mother's education of less than middle school), 26 for the second lowest (less than high-school), and 35 for high SES students (no difference for the top two categories).

²⁶ In the context of Fack et al. (2019), the endogeneity of the cutoffs is driven by the assignment mechanism whereas in this context, the cutoffs are a choice variable of the school. This could be important for the identification of the vertical component in the utility function. I discuss this issue further in the identification section

²⁷ In order to do this, I allow students to rank every school in the market according to their preferences and assign students to schools based on the assumption of preference for higher types over lower types. For more details, see Section 8.

presented here interprets the observed lowest SES as a threshold. A student of type θ_i , which is defined as the mother's education, will choose a school that maximizes the utility among the set of schools with $\theta_{jt}^* \leq \theta_i$.

If we assume that ε_{ijt} is distributed type I extreme value, this produces a logit functional form for the probability that student i chooses school j .

$$P_{ijt} = P(j|\theta^*, \xi, W_i, \mathbf{X}, p, d) = \mathbb{1}(\theta_i > \theta_j^*) \frac{\exp(v_{ijt})}{1 + \sum_{k \in J(\theta_i)} \exp(v_{ikt})}$$

where

$$v_{ijt} = \alpha_i p_{ijt} + \mathbf{X}_{jt-1} \beta_i - \gamma_i d_{ijt} + \lambda_i (\theta_i - \bar{\theta}_{jt-1}) + \xi_{jt}$$

and $J(\theta_i)$ is the choice set of schools available for a student of type θ_i .

Since only differences in utility matter, it is necessary to normalize the utility for one alternative to zero. Effectively, there is no outside option because all students are required to enroll in a school and I observe the entire market. Most schools in each market are quite small, so instead of just normalizing the utility with respect to one particular school, I use the set of public schools in each market as the outside option and normalize the utility for all public schools. I assume that they share the same unobserved quality term (ξ_{jt}) which is normalized to zero and that all public schools are available to everyone. In practice, I am treating public schools as one option, mainly because substitution between public schools is of second order for the mechanisms that explain socioeconomic stratification and this considerably reduces the choice set size.

Probabilities P_{ijt} are conditional on the vector of θ^* . Let $\delta_{jt} = \bar{\beta} X_{jt} + \xi_{jt}$ the year-school specific term that does not vary across students, and $\eta = [\alpha^o, \beta^o, \gamma^o, \lambda^o, \delta]$ the set of parameters to estimate. To recover the parameters from the utility function, I consider the maximum likelihood estimator assuming the observed vector of θ^* :

$$\hat{\eta} = \arg \max L(\eta, \theta^*),$$

where

$$L = \sum_{i=1}^I \sum_{j=1}^J x_{ij} \log(P_{ij})$$

where $x_{ij} = 1$ if student i chooses school j and 0 otherwise.

For the estimation I proceed in two steps. First, separately for each market, I obtain $\alpha^o, \gamma^o, \lambda^o$, and β^o that maximize L , following [Berry \(1994\)](#), I estimate δ_{jt} (year-school specific term) matching the observed market shares for each school to the estimated shares as a function of the parameters in each iteration. This way, the number of parameters to search is significantly reduced, and δ_{jt} allows the model to perfectly match school-level market shares. There are 28 markets and seven years, from 2006 to 2012. The capital city of Santiago is excluded from this part of the analysis because of computational reasons, with over 1000 schools, the choice set is too large, making the estimation infeasible. All other large cities are included, with the largest market outside of Santiago including 230 schools.²⁸

In a second step, from the panel of $\hat{\delta}_{jt}$ and X_{jt} , I estimate the average utility parameters $\bar{\beta}$ from an OLS regression using school fixed effects to control for ξ_j , time invariant unobservable school-level characteristics that may be correlated with X_{jt} . In this step, I pool together all schools across markets.

6.5. Identification

The parameters $\alpha^o, \gamma^o, \beta^o, \lambda^o$ can be identified conditional on the school-year specific term (the vector δ_{jt}). Identification of these parameters is provided by the variation within markets of different types of

students and the schools they enroll. Key variation for identification is given by the changes in the choice set for each type of student before and after the reform and the changes in school characteristics and peer composition. There are two important identification assumptions. First, that parents have correct information regarding feasible schools and they take θ^* as given, analogous to a price taking assumption. In other words, I assume that markets are large enough that the decision of each parent cannot have an effect on admissions thresholds, which is reasonable given the sizes of the markets considered. Second, that the vector δ_{jt} is common across student types. Schools with high cutoffs are by definition only available to high-SES students. Therefore, only high-SES students would contribute to identifying their school-year specific unobserved quality, yet their choice sets contain all schools. Given this, an important identification assumption is that high-SES and low-SES students have the same unobserved quality for each school and only differ in their taste for observed characteristics.

Regarding the choice sets, one concern of the approach presented here is the interpretation of the observed lowest SES as a threshold whereas if schools were not selective, it would merely reflect poor parents not choosing these schools. Nevertheless, including in the choice set schools that are not feasible (the usual approach in the literature when not considering schools' selectivity) is considerably more problematic for correctly recovering preference parameters than not including schools that were in the choice set but not chosen (possible issue in this case).²⁹ Furthermore, the evidence presented in Section 5 points to the fact that many poor students do in fact change their preferred school after the implementation of the reform even in schools without tuition, suggesting that at least some schools were selecting them out (choice sets are not policy invariant). This does not mean that all schools are assumed to be equally selective as thresholds for each school-year adjust flexibly given the observed student allocation in the data. It is important to notice that I am not assuming that parents have complete information over these thresholds. It is very possible that most parents may apply to schools that are infeasible. As applications are not observed, this information cannot be used to recover preference parameters. Still, observing the final enrollment and equilibrium admission thresholds allow the recovery of preferences when including only feasible schools in students' choice sets. I am only ruling out the possibility that parents do not apply to a feasible school because they think it is infeasible (which is also the assumption made when choice sets are not restricted, assuming students could enroll in any school).

The variation that identifies $\bar{\beta}$ comes from the within-school changes in characteristics generated by the SEP reform. I assume that ξ_{jt} is composed by a time-invariant ξ_j and a year-specific idiosyncratic shock $\Delta \xi_{jt}$. A feature of the SEP reform is that it induces variation in the distribution of student types, class size, and test scores within schools, explained by the changes in enrollment following the implementation of the reform. For this reason, it is possible to include both test scores and peer composition in the utility function. Given the within-school variation in the data, I am able to identify $\bar{\beta}$ using school fixed-effects and lagged observable characteristics. The identifying assumption is that changes in unobserved quality are uncorrelated with changes in lagged observed characteristics. In other words, characteristics in X_{jt-1} could be determined based on information on ξ_j but not the idiosyncratic shock $\Delta \xi_{jt}$.

7. Results

Estimates for the average utility parameters are shown in Panel A of [Table 5](#). I estimate the model both with and without cream-skimming restrictions in admissions. The first column shows the results of the full

²⁹ See [Crawford et al. \(2021\)](#) for a more detailed explanation on this result from the literature of consideration sets.

²⁸ For more information regarding the definition of markets see Section 3.1.

Table 5

Average utility parameters.

Panel A: Parameter estimates				
	Dep variable: Average utility (δ_{jt})			
	Restricted choice	Unrestricted choice		
Class Size	−0.011*** (0.002)	−0.005** (0.002)		
Average math score	0.007 (0.031)	−0.084*** (0.029)		
Average peer type	−0.619*** (0.054)	−1.141*** (0.051)		
IQR peer type	−0.121*** (0.026)	−0.095*** (0.025)		
School FE	X	X		
R-squared	0.987	0.989		
N	7179	7179		
Panel B: Correlations across school characteristics				
	Avg. Math score	Avg. Peer type	IQR peer type	Class size
Average math score	1,00			
Average peer type	0,74	1,00		
IQR peer type	−0,32	−0,44	1,00	
Class size	0,02	−0,02	−0,05	1,00

Note: Panel A presents regression coefficients of average utility of a school on different lagged school characteristics: average class size, average standardized mother's education, interquartile range of mother's education, average standardized math test score, and an indicator for participating in the reform. It shows estimates for both, the model including the restriction on the choice set and for the unrestricted version. *, **, and *** denotes significance at the 10, 5 and 1% level. Standard errors are presented in parenthesis. Panel B shows the correlations for school characteristics used in the utility specification.

model including the admission restrictions under which students have limited number of schools available depending on their socioeconomic type. The second column shows results for the unrestricted case, in which case I assume that all schools in a given market are available to all students (the usual assumption in the literature). Differences between Columns 1 and 2 suggest that ignoring admission restrictions leads to underestimating low-SES parents' preferences for school quality (by approximately a factor of two, in terms of test scores and class size) and overestimating their taste for lower type peers. For example, Column 2 shows that parents with low education (the omitted category in the parent education group) have a taste for schools with worse results in standardized tests (negative and significant sign for average math scores). In other words, the model rationalizes the enrollment decisions in the data. If we ignore the restricted choice set and observe low-SES parents not selecting schools with high test scores, one might infer that they have low preferences for school quality. Column 1 shows that this is in fact not true which explains the changes in enrollment of low-SES students following the reform when participating schools start accepting more low-SES students. Panel B of Table 5 shows the correlations across different schools characteristics to have a sense of the variation across these variables.

Table 6 shows estimates for the heterogeneity parameters α^o , γ^o , λ^o , and β^o using the model with restricted choice sets. It shows weighted average coefficients by market size for income and education levels. My results suggest that for parents in the upper categories of education, the most important consideration when choosing a school is the average type of student in the school, and that the magnitude of this parameter increases significantly with the level of income. The coefficient on distance is negative and significant for all groups except the lowest, and of similar magnitudes across groups. Distance is measured in kilometers, tuition in dollars per month, and mother's education and test scores are standardized to have mean zero and standard deviation of one. Therefore, we can interpret the coefficients on average peer type in terms of willingness to travel for a one standard deviation in peer quality. Students with a mother of at most high school education are willing to travel approximately 0.6 km, compared to 3.5 km and 4.9 km

for the groups of more than high school and university education, respectively. The coefficient on tuition is negative and significant only for the lowest education group, which is the only group that experiences significant variation in tuition because they are the group targeted by the SEP reform.

Another interesting result are the coefficients on the difference between own type and average peer type. It shows that for the lowest and highest groups in terms of mother's education, there is a negative taste for schools where the average type is very different from their own type. This does not seem to be the case for the intermediate groups.

Comparing these results with preference estimates in the literature is not straightforward given the different specification for the utility function, in particular due to the restriction in choice sets and the inclusion of peer characteristics. Nevertheless, results are generally in line with related work. Neilson (2021) and Sánchez (2018) estimate demand parameters using similar data. The parameters related to distance sensitivity are in the same order of magnitude, but smaller. This is expected given that they have more accurate student locations.³⁰ Neilson (2021) finds larger heterogeneity in the distance parameter across socioeconomic groups, but this could be related to the different assumptions regarding students' choice sets. Preference parameters on test scores in Neilson (2021) are also positive but not significant, with magnitudes increasing in socioeconomic levels. Sánchez (2018) finds larger preferences for quality, though given the correlation between peer characteristics and test scores, these differences could be explained by the large preferences I find regarding peer quality.

When analyzing these results, it is important to keep in mind that none of the school characteristics, including peer quality, are randomly assigned to schools and these estimates may not capture true preferences if other unobservable characteristics are correlated with peer quality or test scores. Nevertheless, there are two points that provide reassurance for the interpretation of these estimates. First, I am able to control for time invariant unobservables because of the inclusion of school fixed-effects. Second, the SEP reform substantially increases resources for schools serving low-income students. Hence, if anything, unobserved changes in school resources would be inversely related to changes in peer quality. Schools not participating in the program do not experience any changes in resources. Increases in resources from the program are directly related to low peer quality. Accordingly, estimated preferences for peer quality should be a lower bound.

One possible concern is that schools may not only restrict admission at the bottom, but may also exclude students at the top as a response to the new policy. If schools have tight capacity constraints, it is possible that in response to the additional resources from the SEP program they exclude some high-SES students to enroll more low-SES students. If this is happening it would bias the demand estimates for peers as the model would rationalize the fact that high-SES students are less likely to choose SEP schools as a distaste for low-SES peers. I have three arguments to attenuate these concerns. First, looking at Fig. 2, there does not seem to be any evidence of a decrease in probabilities for high-SES students that match the timing and magnitude of the increase for low-SES students. Second, demographic trends in the country have decreased cohort size by 15% from 2004 to 2012, therefore most schools have excess capacity. On average, only 3% of schools reach at least 90% of their observed maximum capacity. Third, given that demand parameters are estimated separately for each market, we can compare estimates for markets in which schools are more crowded with markets in which schools are likely to have excess capacity. If the estimates for the parameter on peers were significantly more heterogeneous and larger in crowded markets, this could reflect a possible bias.

³⁰ School locations are exactly known, but students' location is only observed at the municipality level. To calculate distance, I use the centroid of the municipality. This generates some measurement error, but given that markets contain several municipalities, this should not be a significant limitation.

Table 6
Heterogeneity on preferences by income and mother's education.

	Type:Mother's education				
	<8 years	High school	More than H-S	University	Income
Avg peer type		0.208 (0.513)	1.027** (0.515)	1.376** (0.573)	0.830*** (0.131)
Avg math score		0.003 (0.377)	0.158 (0.375)	0.251 (0.416)	0.097 (0.104)
Class size		0.012 (0.018)	0.019 (0.018)	0.019 (0.021)	-0.003 (0.005)
IQR peer type		0.052 (0.392)	-0.027 (0.389)	0.028 (0.419)	0.031 (0.091)
(Own type-avg type)	-0.204* (0.105)	0.093* (0.055)	0.133** (0.063)	-0.295* (0.156)	-0.018 (0.018)
Distance	0.014 (0.586)	-0.306*** (0.019)	-0.291*** (0.013)	-0.281*** (0.019)	0.016 (0.011)
Tuition	-0.341*** (0.059)	0.013 (0.011)	0.010 (0.008)	0.012 (0.009)	0.014*** (0.003)

Note: This table presents average heterogeneity coefficients across markets weighted by market size. The model allows for heterogeneity in preferences depending on income and mother's education. Income is measured as a continues variable and mother's education as an indicator for being in one of four groups. School characteristics included are average class size, average standardized mother's education, interquartile range of mother's education, average standardized math test score, the distance between the school and the student's municipality measured in kilometers, tuition measured in dollars per month, and the difference between the student's mother's education and the average in the school. *, **, and *** denotes significance at the 10, 5 and 1% level. Standard errors are presented in parenthesis. The estimates shown in this table correspond to the full model that includes the restriction on the choice sets depending on the student type.

Table 7
Heterogeneity in peer preferences by level of market crowdedness.

Willingness to travel for a 1 Std. Dev. Increase in Avg. Peer type			
	Type of student: Mother's education		
	High school	More than H-S	University
Type of market			
Crowded	0.34	3.57	4.83
Semi-crowded	-0.73	1.48	2.72
Not crowded	0.92	6.92	9.29

Note: This table presents average willingness to travel (coefficient on average peer type divided by coefficient on distance) depending on mother's education for each group of markets. The crowdedness of the market is determined by the percentage of private SEP schools that fill at least 90% of the maximum enrollment observed in that school over the time period included in the data. Crowded markets have more than 7% of schools, semi-crowded have between 3 and 7% of schools, and not-crowded have less than 3% of schools.

Given that the relevant set of schools are the private SEP, I classify markets in three groups depending on the percentage of private SEP schools that fill at least 90% of the observed maximum capacity. Not crowded markets are those in which less than 3% of private SEP schools fill at least 90% of their capacity. In semi-crowded markets between 3% and 7%, and in crowded markets more than 7%. These numbers were chosen to have groups of approximately similar number of schools, but results are robust to different limits. As we cannot directly compare coefficients for different markets, Table 7 reports the willingness to travel for a one standard deviation increase in average peer type for each group. We see significant heterogeneity in all three groups. Appendix Table A.1 shows the complete set of estimates for each group of markets.

In summary, these estimates show that the highest heterogeneity in preferences across groups is found in the coefficients on peer quality which would explain some of the patterns described in Section 4, with high-SES students gradually started preferring non-SEP schools after the implementation of the SEP reform when participating schools start enrolling more low-SES students. In the next section, I use these estimates to simulate several counterfactuals to isolate the effects of the different segregation mechanisms.

8. Counterfactuals: Segregation mechanisms

I use the estimated demand parameters to quantify how much of the observed segregation in the data can be attributed to heterogeneity in

parental preferences and how much to school cream-skimming behavior. These are partial equilibrium counterfactuals as in both exercises I assume that schools cannot increase capacity beyond the maximum observed enrollment over the data period and I do not account for potential supply-side responses in price or other characteristics.³¹

Simulating market outcomes in counterfactual experiments requires computing equilibrium admission thresholds. For this, an allocation mechanism has to be defined to obtain the new thresholds. In a first exercise, I assume a centralized admission system to effectively eliminate all cream-skimming behavior from schools. To do this, I allow students to rank all schools and then assume a random serial dictatorship mechanism for assignment. In a second exercise, I do the same, but giving priority to SEP students, to match a reform implemented in Chile in 2017. For each case, I compute the average mother's education in private and public schools and the share of students going to public schools. I also compute actual values from 2012 for comparison.³²

Next, I change preference parameters to eliminate all preferences for peer quality. In this case, to do the matching, I assume that school preferences over students are known and given by the admission model from Section 6. I assume private subsidized schools maintain their admission policy of preferring higher types to lower types and allow admission thresholds to adjust given the new preferences. This generates new equilibrium thresholds.

While each of these counterfactual is computed separately for each market, results are similar across markets. Table 8 shows the results for the largest market in the estimation. The first two columns show the results for eliminating school selection via a centralized lottery, and the third column shows the case of eliminating preference for peer characteristics. Eliminating school selection decreases enrollment in public schools by 2.5 percentage points, and only reduces the gap in mother's education by 30% compared to 2012 (depending on the market, this number goes from 25% to 43%). Giving priority to vulnerable students in the admission lottery marginally reduces the gap with negligible effects on the share for public schools. On the other hand,

³¹ Allowing schools to increase capacity do not qualitatively change the results on segregation, but exaggerates the negative effect over enrollment in public schools. On the other hand, the effect of price adjustments would depend on the specific scenario. More detailed discussion is provided in the main text below.

³² These values do not match the magnitudes from the descriptive analysis because mother's education is standardized for the estimation.

Table 8
Counterfactual comparisons.

	Lottery for school selection	Lottery with priority for low-SES	No Pref. for peer quality	Actual 2012
Avg M.Ed private	0.2815	0.2742	0.2676	0.3647
Avg M.Ed public	-0.3041	-0.2881	-0.1659	-0.4642
Share public schools	0.2589	0.2585	0.3180	0.2835

Note: This table presents simulation results for one representative market assuming different scenarios: first two rows show the average mother's education in public and private school. Mother's education is standardized to have mean zero and standard deviation of one. The third row shows the share of students in public schools when there is a centralized lottery for admission, eliminating selection from schools. Column (1) gives the same priority to everyone, whereas column (2) gives higher priority to vulnerable students. Column (3) shows these same statistics for a hypothetical scenario where parents have no preference for peer characteristics, assuming that schools selection mechanism does not change. Column (4) shows the actual values for 2012 as a comparison.

shutting down parental preferences for peers, increases enrollment in public schools by 3.5 percentage points compared to 2012, and the gap on average mother's education between public and private schools, decreases by almost 50% (depending on the market, this number goes from 42% to 85%). These results show that policies that introduce tiered vouchers are ineffective in reducing socioeconomic segregation when there are large and heterogeneous preferences for peers.

These counterfactuals only simulate short-term responses considering school characteristics from 2012. After the first year, the simulations become significantly more complicated because changing student allocations in one year mechanically changes schools' characteristics with respect to peers, test scores, and class size. Therefore, estimating choices year by year would require at least a model to predict changes in test scores. Having said that, one year should be the correct horizon to study changes in student allocation if we want to abstract from indirect effects through changes in school characteristics. The changes simulated in these counterfactuals reflect the responses that are directly related to school selectivity and taste for peers.

Additionally, it should be noted that these consider only demand-side responses. In the first exercise which eliminates school selection, schools are not allowed to adjust prices. The effect of price adjustments would depend on whether schools are allowed to charge positive prices to all students. Price adjustments would likely result on an attenuation of the observed effects as schools, if allowed, would try to exclude students through price increases.

Similarly, the second exercise which shuts down parental preferences for peers does not allow for possible responses to these changes from schools. Schools could react by modifying the admission process and/or changing prices. There are several possible reasons for schools preferring high-type students: lower marginal cost, position in school rankings, or increased enrollment. The model is agnostic about the reasons for schools engaging in cream-skimming behavior. Schools' preferences for higher types may very well change in response to the change in student preferences. Regarding price adjustments, in this scenario prices would likely decrease as demand for private schools goes down which would amplify the observed effects on segregation. In these counterfactual simulations, I assume that this is not the case in order to isolate the demand side effect on segregation. In this sense, these results represent a lower bound on the total effect that would be amplified if schools respond to the change in preferences by relaxing admission restrictions and decreasing prices.

9. Summary and conclusions

This paper studies the mechanisms behind school segregation using the variation generated by a reform to the Chilean school voucher system. The reform intervened in the educational system in an innovative way that makes it useful to study cream-skimming behavior from private schools. The within-school variation in peer composition, class size, and admission thresholds allows me to estimate parental preferences for school and peer characteristics.

My main results can be summarized in three points. First, I show that private subsidized schools effectively cream-skimmed students

based on socioeconomic characteristics and this behavior decreased following the implementation of the program. Second, estimates for parental preferences differ when accounting for supply-side selection in admissions. Ignoring these restrictions leads to underestimating preferences for school quality. My estimates of structural parameters for parental preferences show that low-SES students care about quality (in terms of test scores and class size) which explains the migration of students from public to private subsidized schools. Third, as parental education increases, so does the magnitude of the coefficients for peer quality. This explains the shift of middle income students from private schools that participated in the program to schools that opted out.

Previous research has suggested that cream-skimming concerns can be alleviated through better program design, for example a tiered voucher system. This paper shows that a tiered voucher does decrease cream-skimming by private subsidized schools which translates in decreased enrollment in public schools. Nevertheless, the results show that this may have little effect on overall stratification given high-SES parents' strong preferences for high-SES peers.

Understanding the role of parental preferences and the mechanisms that underlie school segregation is crucial to evaluate the potential impact of school choice programs on social stratification in schools. School socioeconomic segregation is particularly important given evidence that it can perpetuate long-term income inequality.

Declaration of competing interest

Hereby I declare that I have no material financial interests or any conflict of interest that relate to the research described in this submission to the Journal of Public Economics.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jpubeco.2024.105176>.

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